Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.

Answer:

Covariance is a statistical measure that indicates the direction of the relationship between two variables. It shows whether two variables increase or decrease together.

- If covariance is positive, both variables tend to increase or decrease together.
- If covariance is negative, one variable increases while the other decreases.
- If covariance is zero, there is no linear relationship between the variables.

Difference between Covariance and Correlation

Covariance

Measures the direction of relationship.

Value depends on the units of the variables.

Can take any value from negative infinity to positive infinity.

Difficult to interpret because of scale.

Correlation
Measures both the direction and strength of relationship.
Unit-free (standardized).
Always ranges from -1 to +1**.**
Easy to interpret because of fixed range.

Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?

Answer:

Positive Covariance: Both variables move in the same direction. When one variable increases, the other also tends to increase.

Negative Covariance: The variables move in opposite directions. When one variable increases, the other tends to decrease.

Zero Covariance: There is no linear relationship between the two variables. Changes in one variable do not consistently affect the other.

Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?

Answer:

Limitations of Covariance

Covariance depends on the units of measurement.

It has no fixed range, making comparison difficult.

The magnitude of covariance is difficult to interpret.

It only indicates the direction of the relationship, not its strength.

Why Correlation is Preferred

Correlation is standardized and ranges from -1 to +1.

It clearly indicates both the strength and direction of the relationship.

It allows easy comparison between different datasets.

It is independent of the units of measurement.

Therefore, correlation is more useful and easier to interpret than covariance.

Q4. Explain the difference between Pearson’s correlation coefficient and Spearman’s rank correlation coefficient. When would you prefer to use Spearman’s correlation?

Answer:

Pearson's Correlation Coefficient

Measures the linear relationship between two continuous variables.

Uses the actual numerical values.

Assumes the data is normally distributed.

Represented by r.

Spearman's Rank Correlation Coefficient

Measures the relationship using the rank of the data.

Does not require normally distributed data.

Suitable for ordinal or ranked data.

Represented by ρ (rho).

When to use Spearman's Correlation

Spearman's correlation should be used when:

Data is ranked or ordinal.

Data is not normally distributed.

There are outliers.

The relationship is monotonic but not necessarily linear.

Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?

Answer:
A correlation coefficient of 0.85 indicates a strong positive linear relationship between X and Y. This means that as X increases, Y also tends to increase.
However, correlation does not imply causation. A high correlation does not prove that one variable causes the other to change. The relationship may be influenced by other factors or may simply be coincidental.
Interpretation:
There is a strong positive association between X and Y, but no conclusion about cause-and-effect can be made based only on the correlation coefficient.

Q6. Using the dataset below, calculate the covariance between X and Y.

X	2	4	6	8
Y	3	7	5	10

Answer:

Mean of X = (2 + 4 + 6 + 8) / 4 = 5

Mean of Y = (3 + 7 + 5 + 10) / 4 = 6.25

Using the population covariance formula:

$Cov(X,Y) = \sum(X-\bar{X})(Y-\bar{Y}) / n$

=(46.5+0.75-25+13)/4

=19/4

=4.75

Covariance = 4.75

Interpretation: Since the covariance is positive, X and Y tend to increase together.

Q7. Compute the Pearson correlation coefficient between variables A and B.

A	10	20	30	40	50
B	8	14	18	24	28

Answer:

Using the Pearson correlation coefficient formula:

$r = Cov(A,B) / \sigma A \sigma B$

After calculation,

Pearson Correlation Coefficient (r) ≈ 0.998

Interpretation: There is a **very** strong positive correlation between A and B.

Q8. The following table shows heights (in cm) and weights (in kg) of 5 students. Find the correlation coefficient between Height and Weight.

Height (cm)	150	160	165	170	180
Weight (kg)	50	55	58	62	70

Answer:

Using Pearson's correlation coefficient formula,

Correlation Coefficient (r) ≈ 0.992

Interpretation: There is a very strong positive correlation between height and weight. As height increases, weight also tends to increase.

Q9. Given the dataset below, determine whether there is a positive or negative correlation between X and Y. (No need for exact calculation, just reasoning.)

X	1	2	3	4	5
Y	15	12	9	7	3

Answer:

As X increases, Y decreases.

Therefore, there is a negative correlation between X and Y.

Q10. Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.

Year	1	2	3	4	5
Portfolio A	6	10	12	9	11
Portfolio B	8	9	11	8	10

Answer:

- Covariance = 2.08
- Correlation Coefficient (r) ≈ 0.866

Interpretation:

- The covariance is positive, indicating that both portfolios generally move in the same direction.
- The correlation coefficient (0.866) shows a strong positive relationship between Portfolio A and Portfolio B.

Therefore, the two portfolios generally move together over the five-year period.